

《数据库概论》实验二 高级SQL 实验报告

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实验环境

操作系统为 windows，采用VS code ide 以及其中的 SQL Tools 插件进行代码编写和运行。MySQL 版本是 8.0.46-winx64，高级程序语言代码部分则采用 C 语言。

实验过程

T1.1

代码部分：

```
1 CREATE PROCEDURE info_product (IN product_name VARCHAR(40))
2     SELECT Customer.customerNo, Customer.customerName, OrderDetail.orderNo,
   OrderDetail.quantity, (OrderDetail.price * OrderDetail.quantity)
3     FROM OrderDetail
4     JOIN Product ON OrderDetail.productNo = Product.productNo
5     JOIN OrderMaster ON OrderMaster.orderNo = OrderDetail.orderNo
6     JOIN Customer ON Customer.customerNo = OrderMaster.customerNo
7     WHERE Product.productName = product_name
8     ORDER BY (OrderDetail.price * OrderDetail.quantity) DESC;
9
10 CALL info_product ('32M DRAM');
```

实验截图：

customerNo	customerName	orderNo	quantity	(OrderDetail.pri...
 Filter...	 Filter...	 Filter...	 Filter...	 Filter...
C20050001	统一股份有限公司	200801090001	5	2500.00
C20070002	世界技术开发公司	200803010001	8	1200.00
C20080001	红度股份有限公司	200801090003	5	650.00
C20050004	五一商厦	200803020001	2	200.00

T1.2

代码部分：

```
1 CREATE PROCEDURE info_employee (IN employee_No CHAR(8))
2     SELECT E2.employeeNo, E2.employeeName, E2.gender, E2.hireDate,
   E2.department
3     FROM Employee AS E1
4     JOIN Employee AS E2 ON E1.department = E2.department
5     WHERE E1.employeeNo = employee_No AND E1.hireDate > E2.hireDate;
6
7 CALL info_employee ('E2008005');
```

实验截图：

employeeNo	employeeName	gender	hireDate	department
a Filter...	a Filter...	a Filter...	a Filter...	a Filter...
E2005001	喻自强	M	1990-02-06 00:00:00	财务科

T2.1

代码部分：

```
1 CREATE FUNCTION average_price(product_name VARCHAR(40))
2 RETURNS FLOAT
3 READS SQL DATA
4 RETURN (
5     SELECT SUM(quantity * price) / SUM(quantity)
6     FROM OrderDetail
7     JOIN Product ON Product.productNo = OrderDetail.productNo
8     WHERE Product.productName = product_name
9 );
10
11 SELECT productName, average_price(productName)
12 FROM Product;
```

实验截图：

productName	average_price(pr...
a Filter...	a Filter...
32M DRAM	227.5
17寸显示器	425
120GB硬盘	215
3.5寸软驱	350
键盘	375
VGA显示卡	337.5
网卡	258.571
Pentium100CPU	290
1G DDR	315.467
52倍速光驱	362.5
计算机字典	286.364
9600bits/s调制解调	342
Pentium主板	245.714
硕泰克SL—K8AN-RL主...	265.714
龙基777FT纯平显示器	215.333

T2.2

代码部分：

```
1 CREATE FUNCTION count_sum (product_no CHAR(9))
2 RETURNS INTEGER
3 READS SQL DATA
4 RETURN (
5     SELECT SUM(quantity)
6     FROM OrderDetail
7     WHERE productNO = product_no
8 );
9
10 SELECT productNo, productName, count_sum(productNo)
11 FROM Product
12 WHERE count_sum(productNo) > 4;
```

实验结果：

productNo ↑	productName	count_sum(prod...
abc Filter...	abc Filter...	abc Filter...
P20050001	32M DRAM	20
P20050003	120GB硬盘	8
P20050004	3.5寸软驱	7
P20050005	键盘	6
P20060002	网卡	7
P20060003	Pentium100CPU	12
P20070001	1G DDR	15
P20070002	52倍速光驱	12
P20070003	计算机字典	11
P20070004	9600bits/s调制解调	5
P20080001	Pentium主板	7
P20080002	硕泰克SL—K8AN-RL主...	7

T3.1

代码部分：

```
1 CREATE TRIGGER set_product_price
2 BEFORE INSERT ON Product
3 FOR EACH ROW
4 BEGIN
5     IF NEW.productPrice > 1000
```

```

6      THEN SET NEW.productPrice = 1000;
7      END IF;
8  END;
9
10 INSERT INTO Product
11 VALUES ('114514', 'tempProduct', 'tempClass', 1919.81);
12
13 SELECT *
14 FROM Product
15 WHERE productName = 'tempProduct';

```

实验结果：

productNo ↑	productName	productClass	productPrice
abc Filter...	abc Filter...	abc Filter...	abc Filter...
114514	tempProduct	tempClass	1000.00

T3.2

代码部分：

```

1  DROP TRIGGER IF EXISTS add_salary;
2
3  CREATE TRIGGER add_salary
4  AFTER INSERT ON OrderMaster
5  FOR EACH ROW
6  BEGIN
7      UPDATE Employee
8      SET salary = salary *
9          CASE
10             WHEN hireDate < '1992-01-01' THEN 1.08
11             ELSE 1.05
12          END
13      WHERE employeeNo = NEW.employeeNo;
14 END;
15
16 SELECT employeeName, salary
17 FROM employee
18 WHERE employeeNo = 'E2005001' OR employeeNo = 'E2005005';
19
20 INSERT INTO OrderMaster
21 VALUES ('123', 'C20050001', 'E2005001', '20260521', 0.00, '1');
22 INSERT INTO OrderMaster
23 VALUES ('456', 'C20050002', 'E2005005', '20060520', 0.00, '2');
24
25 SELECT employeeName, salary
26 FROM employee
27 WHERE employeeNo = 'E2005001' OR employeeNo = 'E2005005';
28
29 DROP TRIGGER IF EXISTS add_salary;

```

实验结果：

加订单前：

employeeName	hireDate	salary
abc Filter...	abc Filter...	abc Filter...
喻自强	1990-02-06 00:00:00	5800.00
张小东	1992-03-28 00:00:00	1800.00

加订单后:

employeeName	hireDate	salary
abc Filter...	abc Filter...	abc Filter...
喻自强	1990-02-06 00:00:00	6264.00
张小东	1992-03-28 00:00:00	1890.00

T4

实验代码:

```

1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <mysql.h>
4
5  int main(void){
6      MYSQL *conn;
7      MYSQL_RES *result;
8      MYSQL_ROW row;
9
10     // MySQL 连接参数, 对应本机上的 OrderDB 数据库。
11     const char *host = "127.0.0.1";
12     const char *user = "lyh";
13     const char *password = "123456";
14     const char *database = "OrderDB";
15     const unsigned int port = 3306;
16
17     conn = mysql_init(NULL);
18     mysql_real_connect(conn, host, user, password, database, port, NULL,
19     0);
20
21     // 查询工资最高的 20 名员工。
22     const char* sql1 =
23         "SELECT employeeNo, employeeName, salary "
24         "FROM Employee "
25         "ORDER BY salary DESC "
26         "LIMIT 20";
27
28     mysql_query(conn, sql1);
29     result = mysql_store_result(conn);
30
31     printf("%-10s %-10s %-10s\n", "employeeNo", "employeeName", "salary");
32     printf("-----\n");
33     while ((row = mysql_fetch_row(result)) != NULL) {
34         printf(

```

```

34         "%-10s %-10s %-10s\n",
35         row[0] ? row[0] : "NULL",
36         row[1] ? row[1] : "NULL",
37         row[2] ? row[2] : "NULL"
38     );
39 }
40
41 // 插入一条客户记录，并再次查询该客户检查插入结果。
42 const char* sql2 =
43     "INSERT INTO Customer "
44     "VALUES ('C20080002', '泰康股份有限公司', '010-5422685', '天津市',
45     '220501')";
46
47 const char* sql2_test =
48     "SELECT * "
49     "FROM Customer "
50     "WHERE customerNo = 'C20080002'";
51
52 mysql_query(conn, sql2);
53 mysql_query(conn, sql2_test);
54 result = mysql_store_result(conn);
55
56 printf("%-10s %-10s %-10s %-10s %-10s\n", "customerNo", "customerName",
57 "telephone", "address", "zip");
58 printf("-----
59 \n");
60 while ((row = mysql_fetch_row(result)) != NULL) {
61     printf(
62         "%-10s %-10s %-10s %-10s %-10s\n",
63         row[0] ? row[0] : "NULL",
64         row[1] ? row[1] : "NULL",
65         row[2] ? row[2] : "NULL",
66         row[3] ? row[3] : "NULL",
67         row[4] ? row[4] : "NULL"
68     );
69 }
70
71 // 删除工资大于 5000 的员工，删除前后各打印一次便于对比。
72 const char* sql3 =
73     "DELETE FROM Employee "
74     "WHERE salary > 5000";
75
76 const char* sql3_test =
77     "SELECT employeeName, salary "
78     "FROM Employee";
79
80 mysql_query(conn, sql3_test);
81 result = mysql_store_result(conn);
82
83 printf("%-10s %-10s\n", "employeeName", "salary");
84 printf("-----
85 \n");
86 while ((row = mysql_fetch_row(result)) != NULL) {
87     printf(
88         "%-10s %-10s\n",
89         row[0] ? row[0] : "NULL",
90         row[1] ? row[1] : "NULL"
91     );
92 }

```

```

88     }
89
90     mysql_query(conn, sql3);
91
92     mysql_query(conn, sql3_test);
93     result = mysql_store_result(conn);
94
95     printf("%-10s %-10s\n", "employeeName", "salary");
96     printf("-----
\n");
97     while ((row = mysql_fetch_row(result)) != NULL) {
98         printf(
99             "%-10s %-10s\n",
100             row[0] ? row[0] : "NULL",
101             row[1] ? row[1] : "NULL"
102         );
103     }
104
105
106     // 将价格大于 1000 的产品打五折，更新前后各打印一次便于对比。
107     const char* sql4 =
108         "UPDATE Product "
109         "SET productPrice = productPrice * 0.5 "
110         "WHERE productPrice > 1000";
111
112     const char* sql4_test =
113         "SELECT productName, productPrice "
114         "FROM Product";
115
116     mysql_query(conn, sql4_test);
117     result = mysql_store_result(conn);
118
119     printf("%-10s %-10s\n", "productName", "productPrice");
120     printf("-----
\n");
121     while ((row = mysql_fetch_row(result)) != NULL) {
122         printf(
123             "%-10s %-10s\n",
124             row[0] ? row[0] : "NULL",
125             row[1] ? row[1] : "NULL"
126         );
127     }
128     mysql_query(conn, sql4);
129     mysql_query(conn, sql4_test);
130     result = mysql_store_result(conn);
131
132     printf("%-10s %-10s\n", "productName", "productPrice");
133     printf("-----
\n");
134     while ((row = mysql_fetch_row(result)) != NULL) {
135         printf(
136             "%-10s %-10s\n",
137             row[0] ? row[0] : "NULL",
138             row[1] ? row[1] : "NULL"
139         );
140     }
141 }

```

实验结果：

其中部分图像中左侧为操作前，右侧为操作后

employeeNo	employeeName	salary
E2005001	喻自强	5800.00
E2008005	张小梅	5000.00
E2005004	张露	4100.00
E2006001	陈辉	4000.00
E2008004	李虹冰	3400.00
E2008001	陈诗杰	3200.00
E2008003	黄梅莹	3100.00
E2008002	张良	2700.00
E2005003	张小娟	2600.00
E2006002	韩梅	2600.00
E2006003	刘风	2500.00
E2007001	吴浮萍	2500.00
E2005002	张小梅	2400.00
E2007002	高代鹏	2000.00
E2005005	张小东	1800.00

customerNo	customerName	telephone	address	zip
C20080002	泰康股份有限公司	010-5422685	天津市	220501

employeeName	salary	employeeName	salary
喻自强	5800.00	张小梅	2400.00
张小梅	2400.00	张小娟	2600.00
张小娟	2600.00	张露	4100.00
张露	4100.00	张小东	1800.00
张小东	1800.00	陈辉	4000.00
陈辉	4000.00	韩梅	2600.00
韩梅	2600.00	刘风	2500.00
刘风	2500.00	吴浮萍	2500.00
吴浮萍	2500.00	高代鹏	2000.00
高代鹏	2000.00	陈诗杰	3200.00
陈诗杰	3200.00	张良	2700.00
张良	2700.00	黄梅莹	3100.00
黄梅莹	3100.00	李虹冰	3400.00
李虹冰	3400.00	张小梅	5000.00
张小梅	5000.00		

productName	productPrice	productName	productPrice
32M DRAM	80.70	32M DRAM	80.70
17寸显示器	700.00	17寸显示器	700.00
120GB硬盘	300.00	120GB硬盘	300.00
3.5寸软驱	35.00	3.5寸软驱	35.00
键盘	100.60	键盘	100.60
VGA显示卡	1200.60	VGA显示卡	600.30
网卡	66.00	网卡	66.00
Pentium100CPU	200.00	Pentium100CPU	200.00
1G DDR	256.00	1G DDR	256.00
52倍速光驱	200.00	52倍速光驱	200.00
计算机字典	100.00	计算机字典	100.00
9600bits/s调制解调	320.00	9600bits/s调制解调	320.00
Pentium主板	890.00	Pentium主板	890.00
硕泰克SL-K8AN-RL主板	1100.00	硕泰克SL-K8AN-RL主板	550.00
龙基777FT纯平显示器	900.00	龙基777FT纯平显示器	900.00

代码部分：

```

1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <mysql.h>
4
5  void print(MYSQL_RES* result){
6
7  }
8
9  int main(void){
10     MYSQL *conn;
11     MYSQL_RES *result;
12     MYSQL_ROW row;
13
14     // MySQL 连接参数, 对应本机上的 OrderDB 数据库。
15     const char *host = "127.0.0.1";
16     const char *user = "lyh";
17     const char *password = "123456";
18     const char *database = "OrderDB";
19     const unsigned int port = 3306;
20
21     conn = mysql_init(NULL);
22     mysql_real_connect(conn, host, user, password, database, port, NULL, 0);
23
24     // 从键盘读入部门名, 用它拼接后面的更新和查询语句。
25     char str_department[50];
26     char sql1[200];
27     char sql1_test[200];
28     scanf("%49s", str_department);
29
30     // 给指定部门的员工统一加薪 200。
31     snprintf(sql1, sizeof(sql1),
32              "UPDATE Employee "
33              "SET salary = salary + 200 "
34              "WHERE department = '%s'", str_department);
35
36     // 查询该部门员工工资, 用于加薪前后对比。
37     snprintf(sql1_test, sizeof(sql1_test),
38              "SELECT employeeName, salary "
39              "From Employee "
40              "WHERE department = '%s'", str_department);
41
42     mysql_query(conn, sql1_test);
43     result = mysql_store_result(conn);
44     printf("%-10s %-10s\n", "employeeName", "salary");
45     printf("-----\n");
46     while ((row = mysql_fetch_row(result)) != NULL) {
47         printf(
48             "%-10s %-10s\n",
49             row[0] ? row[0] : "NULL",
50             row[1] ? row[1] : "NULL"
51         );
52     }

```

```

53     mysql_query(conn, sql1);
54     mysql_query(conn, sql1_test);
55     result = mysql_store_result(conn);
56     printf("%-10s %-10s\n", "employeeName", "salary");
57     printf("-----
\n");
58     while ((row = mysql_fetch_row(result)) != NULL) {
59         printf(
60             "%-10s %-10s\n",
61             row[0] ? row[0] : "NULL",
62             row[1] ? row[1] : "NULL"
63         );
64     }
65
66     // 查询所有客户的名称、地址和电话。
67     const char* sql2 =
68         "SELECT customerName, address, telephone "
69         "FROM Customer";
70
71     mysql_query(conn, sql2);
72     result = mysql_store_result(conn);
73     printf("%-10s %-10s %-10s\n", "customerName", "address", "telephone");
74     printf("-----
\n");
75     while ((row = mysql_fetch_row(result)) != NULL) {
76         printf(
77             "%-10s %-10s %-10s\n",
78             row[0] ? row[0] : "NULL",
79             row[1] ? row[1] : "NULL",
80             row[2] ? row[2] : "NULL"
81         );
82     }
83 }

```

实验结果：

其中部分图像中左侧为操作前，右侧为操作后

employeeName	salary	employeeName	salary
张小梅	2400.00	张小梅	2600.00
张小娟	2600.00	张小娟	2800.00
张露	4100.00	张露	4300.00
张小东	1800.00	张小东	2000.00
韩梅	2600.00	韩梅	2800.00
刘风	2500.00	刘风	2700.00
吴浮萍	2500.00	吴浮萍	2700.00
张良	2700.00	张良	2900.00
黄梅莹	3100.00	黄梅莹	3300.00
李虹冰	3400.00	李虹冰	3600.00

customerName	address	telephone
统一股份有限公司	天津市	022-3566021
兴隆股份有限公司	天津市	022-3562452
上海生物研究室	北京市	010-2121000
五一商厦	上海市	021-4532187
大地商城	北京市	010-1165152
联合股份有限公司	上海市	021-4568451
南昌市电脑研制中心	南昌市	0791-4412152
世界技术开发公司	上海市	021-4564512
万事达股份有限公司	天津市	022-4533141
红度股份有限公司	北京市	010-5421585

实验中遇到的困难及解决办法

参考文献及致谢
